i. **Dermatogen:** It is the outermost layer. It gives rise to root epidermis.
ii. **Periblem:** It is the middle layer. It gives rise to cortex**.**
iii. **Plerome:** It is innermost layer. It gives rise to stele
iv. **Calyptrogen:** It gives rise to root cap.

Korper Kappe Theory

Korper Kappe theory is proposed by **Schuepp**. There are two zones in root apex – Korper and Kappe

1. **Korper zone** forms the body.
2. **Kappe zone** forms the cap. This theory is equivalent to tunica corpus theory of shoot apex.The two divisions are distinguished by the type of T (also called Y) divisions. Korper is characterised by inverted T divisions and kappe by straight T divisions.

Quiescent Centre Concept

Quiescent centre concept was proposed by **Clowes** (1961) to explain root apical meristem activity. This centre is located between root cap and differentiating cells of the roots. The apparently inactive region of cells in root promeristem is called quiescent centre. It is the site of hormone synthesis and also the ultimate source of all meristematic cells of the meristem.

9.2 Permanent Tissues

The Permanent tissues develop from apical meristem. They lose the power of cell division either permanently or temporarily. They are classified into two types:

1. Simple permanent tissues.
2. Complex permanent tissues.

Simple Permanent Tissues

Simple tissues are composed of one type of cells only. The cells are structurally and functionally similar. It is of three types.

1. Parenchyma
2. Collenchyma
3. Sclerenchyma

Parenchyma (Gk: *Para*-beside; *enehein*- to pour)

Parenchyma is generally present in all organs of the plant. It forms the ground tissue in a plant. Parenchyma is a living tissue and made up of thin walled cells. The cell wall is made up of cellulose. Parenchyma cells may be oval, polyhedral, cylindrical, irregular, elongated or armed. The tissue normally has prominent intercellular spaces and may store various types of materials like, water, air, ergastic substances.

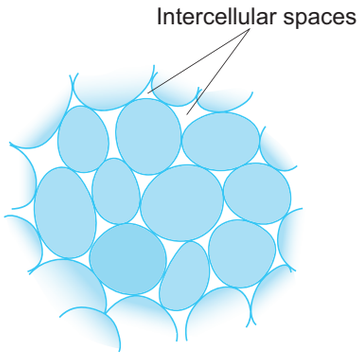


Figure 9.4: Parenchyma

Occsionally Parenchyma cells which store resin, tannins, crystals of calcium carbonate, calcium oxalate are called idioblasts. Parenchyma is of different types and some of them are discussed as follows.

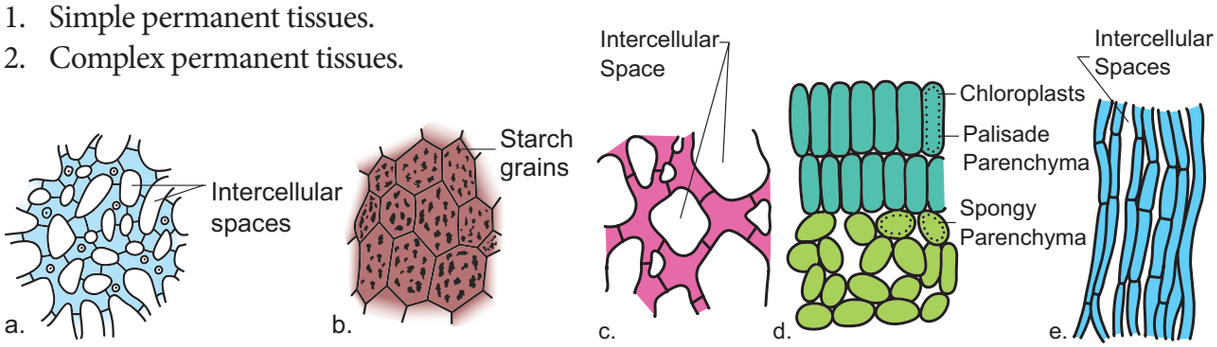


Figure 9.5: Types of Parenchyma
a) Aerenchyma, b) Storage parenchyma c) Stellate parenchyma, d) Chlorenchyma, e) Prosenchyma

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